

i When to Use

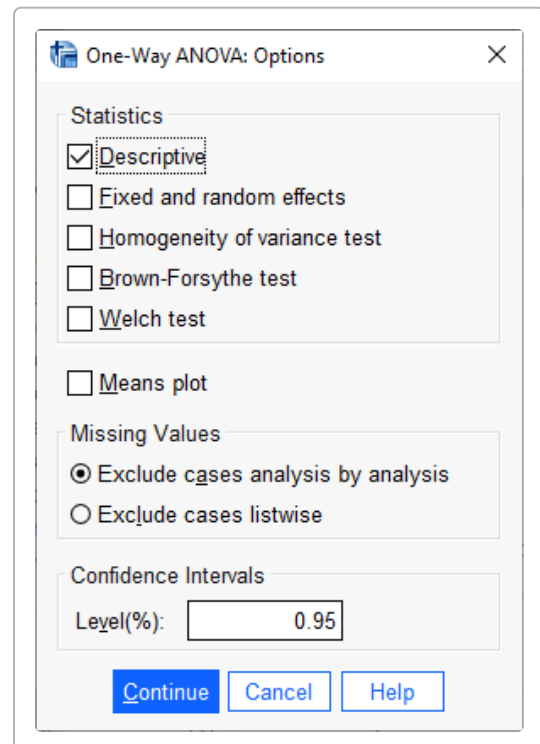
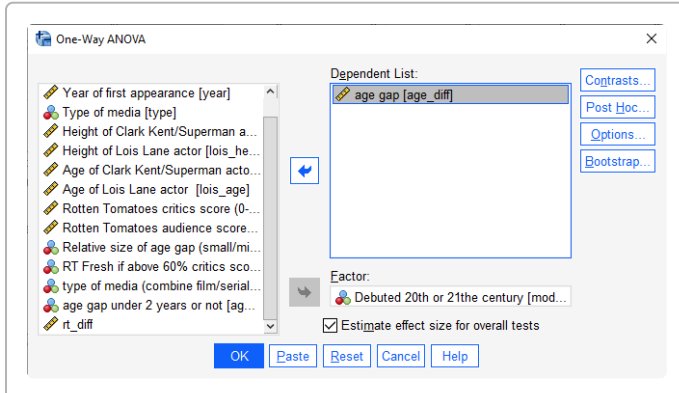
To compare means of a quantitative variable obtained from 2 or more independent groups.

H Research Hypothesis: There are significant differences in emotional impact scores among the three age difference categories (Minimal, Average, Big), with portrayals featuring larger age differences receiving higher emotional impact ratings.

H Null Hypothesis: The three different age difference categories have the same mean emotional impact score.

Analyze → General Linear Model → Univariate

- Highlight the “Dependent” variable (be sure it is **quantitative**) and click the arrow
- Highlight the “Factor” (IV, grouping) variable (be sure it is **qualitative**) and click the arrow
- “Options” — check that you want “Descriptive Statistics”
- “Post Hoc” — highlight the factor and click the arrow



Note

You can also perform this analysis using the “ONEWAY” procedure we used for the 2 BG ANOVA. It has the same post hoc analysis choices and produces equivalent output.

```
UNIANOVA emotional_impact BY age_grp /METHOD=SSTYPE(3) /POSTHOC=age_grp(TUKEY LSD)
/PRINT=DESCRIPTIVE /CRITERIA=ALPHA(.05).
```

- ① DV “by” IV
- ② Appropriate for =/n
- ③ IV & enter Post Hocs
- ④ Get descriptive stats
- ⑤ Sets p-value for the Post hoc tests

SPSS Output

Descriptives

emotional_impact							
				95% Confidence Interval for Mean			
	N	Mean	Std. Deviation	Std. Error	Lower Bound	Upper Bound	Minimum Maximum
Minimal	12	13.08	3.00	0.87	11.18	14.99	8.00 18.00
Average	16	14.62	2.92	0.73	13.07	16.18	10.00 19.00
Big	18	12.39	3.73	0.88	10.53	14.24	4.00 18.00
Total	46	13.35	3.35	0.49	12.35	14.34	4.00 19.00

ANOVA

emotional_impact					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	43.490	2	21.745	2.020	.145
Within Groups	462.944	43	10.766		
Total	506.435	45			

Multiple Comparisons

Multiple Comparisons

Dependent Variable: emotional_impact — LSD

(I) age_grp	(J) age_grp	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	Upper Bound
Minimal	Average	-1.542	1.253	.225	-4.069	0.985
Minimal	Big	0.694	1.223	.573	-1.772	3.161
Average	Minimal	1.542	1.253	.225	-0.985	4.069
Average	Big	2.236	1.127	.054	-0.037	4.510
Big	Minimal	-0.694	1.223	.573	-3.161	1.772
Big	Average	-2.236	1.127	.054	-4.510	0.037

Assuming the null hypothesis is true, the p-value is the probability of obtaining an F statistic at least as large as the one observed. It is not the probability that the result is a Type I error.

SPSS may display **.000** when a very small p-value rounds to three decimal places. Report that value as **p < .001**, not p = .000.

Notice that each pairwise comparison is presented twice!

Minimal vs Average = Average vs Minimal

Be sure you get the direction of each significant mean difference right!!

Minimal = Average

Minimal = Big

Average = Big

These LSD p-values can also be used for Bonferroni tests. Had we been interested in only the comparison of Minimal v Average & Average v Big, we would want to test each using $p = .05 / 2 = .025$.

Multiple Comparisons

Dependent Variable: emotional_impact — Tukey HSD

(I) age_grp	(J) age_grp	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Minimal	Average	-1.542	1.253	.442	-4.583	1.500
Average	Minimal	1.542	1.253	.442	-1.500	4.583
Minimal	Big	0.694	1.223	.838	-2.274	3.663
Big	Minimal	-0.694	1.223	.838	-3.663	2.274
Average	Big	2.236	1.127	.129	-0.501	4.973
Big	Average	-2.236	1.127	.129	-4.973	0.501

Notice that each pairwise comparison is presented twice!

Minimal vs Average = Average vs Minimal

Be sure you get the direction of each significant mean difference right!!

Minimal = Average

Minimal = Big

Average = Big

Usually, but not always, results from LSD & HSD analyses agree. When they don't, the source of the disagreement is usually that the study is "underpowered" for the HSD. A sample size providing adequate power to reject H0 using the more sensitive LSD may not provide adequate power to reject H0 using the more conservative HSD. This most likely to happen when the sample size is selected based on $p < .05$, and then the HSD is applied.

Post Hoc "Computators"

SPSS does not provide post hoc analyses for all ANOVA models (e.g., WG designs). Also, there may be occasions when you want to compare means from a study that didn't post analyses, or did them differently than you would have preferred. One additional advantage of using these is that you can provide your readers with the LSD or HSD values that were the basis of your post hoc tests.

Balanced-design formulas (equal group sizes)

Use these formulas when **every group has the same sample size**, denoted by n . Because the same n applies to every group, one LSD threshold and one HSD threshold can be compared with every pairwise mean difference.

LSD minimum mean difference

1. Determine $df(\text{error})$ for the analysis.
2. Use the t-table to determine the critical value of t for $df(\text{error})$.
3. Determine MS_{Error} from the ANOVA table.
4. Determine the common sample size, n , in each group.
5. Compute one LSD minimum mean difference:

$$d_{LSD} = t_{critical} \times \sqrt{\frac{2 \times MS_{Error}}{n}}$$

HSD minimum mean difference

1. Determine the number of conditions, k .
2. Determine $df(\text{error})$.
3. Use the q-table to determine the critical studentized-range value for k and $df(\text{error})$.
4. Determine MS_{Error} and the common sample size, n .
5. Compute one HSD minimum mean difference:

$$d_{HSD} = q_{critical} \times \sqrt{\frac{MS_{Error}}{n}}$$

Note

The balanced formulas are the equal-group-size simplifications of the formulas below. Do not substitute an average group size when the observed group sizes differ.

Extra: Unequal group sizes

This example is unbalanced: the Minimal, Average, and Big groups contain 12, 16, 18 observations. Each comparison therefore has a different standard error and a different minimum mean difference. Use the two actual sample sizes for the groups in each comparison.

Unequal-group LSD minimum mean differences

1. Determine df(error) for the analysis: **df(error) = 43**
2. Use the t-table to determine the critical value of t (at p = .05) for this df(error): **t-critical = 2.017**
3. Determine the MSerror for the analysis: **MSerror = 10.766**
4. For each comparison, use the sample sizes of the two groups being compared.
5. Apply the unequal-group LSD formula to obtain a pair-specific minimum mean difference:

$$d_{LSD,ij} = t_{critical} \times \sqrt{MS_{Error} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

Comparison	n1	n2	LSD minimum difference
Minimal v Average	12	16	2.527
Minimal v Big	12	18	2.466
Average v Big	16	18	2.274

Unequal-group Tukey–Kramer minimum mean differences

1. Determine the number of conditions: **k = 3**
2. Determine df(error): **df(error) = 43**
3. Use the q-table (Studentized Range Statistic) for k and df(error): **q-critical = 3.433**
4. Determine the MSerror: **MSerror = 10.766**
5. For each comparison, use the sample sizes of the two groups being compared.
6. Apply the Tukey-Kramer formula:

$$d_{TK,ij} = q_{critical} \times \sqrt{\frac{MS_{Error}}{2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

Comparison	n1	n2	Tukey-Kramer minimum difference
Minimal v Average	12	16	3.042
Minimal v Big	12	18	2.968
Average v Big	16	18	2.737

Compare each mean difference with its matching pair-specific threshold:

- If the mean difference is **larger** than its threshold, the two means are statistically significantly different.
- If it is **smaller**, the comparison is not statistically significant. This does not establish equivalence; equivalence requires a separate equivalence test and a prespecified equivalence bound.

Applying LSD/HSD values to the pairwise comparisons

Applying these LSD/HSD values to the pairwise comparisons... Minimal = 13.08 Average = 14.62 Big = 12.39

	Minimal v Average	Minimal v Big	Average v Big
Mean Difference	1.54	0.69	2.24
LSD Result	=	=	=
HSD Result	=	=	=
RH:	<	<	<
RH: support?	Not supported	Not supported	Not supported

Reporting the Results

Sample APA Write-Up

The emotional impact scores for each age difference category are summarized in the table above. Minimal ($M = 13.08$, $SD = 3.00$), Average ($M = 14.62$, $SD = 2.92$), Big ($M = 12.39$, $SD = 3.73$). There were no significant mean differences in emotional impact among the three age difference categories, $F(2,43) = 2.02$, $MSe = 10.77$, $p = .145$. Pairwise comparisons using LSD with the pair-specific minimum mean differences shown above revealed that there was no difference between the average emotional impact of Minimal and Average portrayals or between the average emotional impact of Minimal and Big portrayals or between the average emotional impact of Average and Big portrayals.

Key Reporting Elements

When reporting k BG ANOVA results with post hocs, be sure to communicate:

- The research hypothesis (if there is one)
- Descriptive statistics (M , SD) for each group
- The overall ANOVA results (F , df , p , MSe)
- The post hoc method used and the pair-specific minimum mean difference values
- The direction of each significant pairwise difference
- Which pairwise comparisons were not significant
- Whether or not results support the research hypothesis